

CARMA: Large-Scale Arabic Reddit Corpus for Mental Health Analysis

Anonymous submission

Abstract

Mental health has emerged as a critical global issue, yet Arabic NLP resources for mental health research remain scarce. We introduce **CARMA**, a large-scale corpus from Reddit (4,365 diagnosed users and 2,875 control, **~625K** posts), covering users who have self-reported diagnoses of mental health conditions. The corpus targets 12 conditions and a control group. We collect data using a precision-oriented, pattern-based extraction pipeline (estimated at ~78% precision by manual annotation) validated through manual human annotation that combines self-disclosure phrases with condition-specific keywords designed to support four Arabic dialect groups (Gulf, Egyptian, Levantine, and Modern Standard Arabic) to identify posts containing genuine diagnostic self-reports, though the resulting corpus skews toward Egyptian and Gulf Arabic rather than representing these groups evenly. We employ an author-level approach that retrieves complete posting histories including relevant meta data (subreddit, time of creation, interaction stats) to enable longitudinal analysis. We benchmark the performance of current detection models and leverage Large Language Models to establish baselines for Arabic mental health classification.

1 Introduction

Mental health has emerged as a critical global challenge, with over one billion people worldwide estimated to live with a mental disorder (World Health Organization 2025). In recent years, social media platforms have become a valuable source of large-scale, naturalistic text for studying mental health-related language. Computational approaches have demonstrated that linguistic and behavioral signals in user-generated content can serve as proxies for detecting mental health conditions (De Choudhury et al. 2013; Chancellor and De Choudhury 2020). The CLPsych workshop series further institutionalized this line of research by establishing datasets, shared tasks and standardized evaluation frameworks (Coppersmith et al. 2015). A central methodological development is the *self-reported diagnosis* paradigm, which identifies users who publicly disclose having received a clinical diagnosis. First introduced by Coppersmith, Dredze, and Harman (2014a) on Twitter and later extended to Reddit by Cohan et al. (2018), this approach has enabled the construction of large-scale datasets without requiring manual clinical annotation.

Despite this progress, the overwhelming majority of existing resources target English-language data (Harrigan, Aguirre, and Dredze 2021). Arabic remains substantially underrepresented in computational mental health research, despite being spoken by over 400 million people across diverse linguistic and cultural contexts. Arabic poses particular challenges for natural language processing, including rich morphology and extensive dialectal variation across regional varieties such as Gulf, Egyptian, and Levantine Arabic (Darwish et al. 2021b; Habash 2010). Existing Arabic mental health datasets are limited in scale, typically target only one or two conditions (most commonly depression or suicidal ideation), and rarely address dialectal variation systematically (see Table 1).

In this paper, we address these gaps by introducing **CARMA**, a large-scale and carefully-curated Arabic mental health corpus collected from Reddit. The corpus follows the self-reported diagnosis paradigm: we develop a dialect-aware, pattern-based extraction pipeline. We adopt an author-level collection strategy, retrieving the complete posting histories of identified users to support longitudinal analysis of linguistic patterns beyond the initial disclosure posts and including further metadata about the post. The corpus targets 12 conditions and a control group (Table 1).¹

Our contributions are as follows:

1. We introduce **CARMA**, a large-scale Arabic mental health corpus collected from Reddit, targeting 12 conditions and a control group.
2. We develop the first precision-oriented, dialect aware, automatic pattern-based extraction pipeline to identify self disclosure in Arabic.
3. We validate the automatic extraction pipeline through manual annotation.
4. We provide a comprehensive analysis of corpus characteristics and baseline model performance, establishing a benchmark for Arabic mental health detection.

¹Throughout this paper, *diagnosed* and *control* are shorthand for *self-reported* group assignment, not clinical verification: “diagnosed” users are those who publicly self-disclosed a diagnosis, and “control” users are those screened for the absence of such disclosure, not clinically confirmed to be healthy. See §3.2 for the full discussion and its limitations.

2 Related Work

2.1 Mental Health Detection from Social Media

Research on detecting mental health conditions from social media has grown substantially over the past decade, from early work demonstrating that linguistic and behavioral signals can identify users experiencing depression (De Choudhury et al. 2013) to the CLPsych workshop series (Coppersmith et al. 2015), which introduced standardized shared tasks and evaluation frameworks that helped establish methodological foundations for the field.

A key development is the *self-reported diagnosis* paradigm, which identifies users who publicly disclose a clinical diagnosis through pattern matching. Coppersmith, Dredze, and Harman (2014a) first introduced this approach on Twitter via regular-expression matching; Cohan et al. (2018) extended it to Reddit with SMHD, a large-scale corpus covering nine conditions, and Mankarious et al. (2025) followed a similar methodology at larger scale on the same platform with MINDSET. The paradigm has since been applied to other languages, including German (SMHD-GER, Zanwar et al. 2023). Harrigian, Aguirre, and Dredze (2021) survey existing social media mental-health datasets, noting that the majority target English, leaving other languages substantially underrepresented.

2.2 Arabic NLP for Mental Health

Compared to English, computational research on mental health in Arabic remains relatively limited. Arabic presents several challenges for natural language processing, including rich morphology, substantial dialectal variation across regional varieties such as Gulf, Egyptian, and Levantine Arabic, and limited availability of large annotated datasets (Darwish et al. 2021a; Alayba 2025). Table 1 summarizes existing Arabic datasets related to mental health and situates our corpus within this landscape.

Several studies have constructed X (Twitter)-based datasets targeting individual conditions: OCD symptoms (Al-Haider et al. 2024), suicidal ideation (Abdulsalam, Al-hothali, and Al-Ghamdi 2024; Baghdadi et al. 2022), and, most prominently, depression (Hassib et al. 2022; Almouzi, Alageel et al. 2019; El-Ramly et al. 2021; Maghraby and Ali 2022), including PHQ-9-organized (Maghraby and Ali 2022) and comorbid depression/suicidal-ideation (Hassib et al. 2022) variants. Beyond social media, Alhuzali, Alasmari, and Alsaleh (2024) introduce MentalQA, a question-answering dataset from the Altibbi medical platform rather than user-generated discourse. Table 1 details dataset sizes, target conditions, and annotation methods.

Despite these efforts, existing Arabic mental health datasets exhibit three notable limitations. First, most datasets focus on only one or two conditions, typically depression or suicidal ideation, leaving the broader spectrum of mental health conditions largely unaddressed. Second, annotations are typically performed at the post level, without modeling user-level context or longitudinal behavior. Third, dialectal variation across Arabic varieties is rarely addressed systematically during dataset construction. Our corpus is designed to address all three of these limitations.

3 Dataset Construction

3.1 Diagnosed Dataset

We collect data from Reddit using pre-downloaded JSONL archives from Arctic Shift² (Pushshift format), containing posts and comments from Arabic-speaking subreddits³. Reddit offers longer, more detailed self-reports that are well-suited to the self-reported diagnosis paradigm. To the best of our knowledge, there are no dedicated Arabic-language mental health subreddits comparable to those in English (e.g., r/depression, r/SuicideWatch). Consequently, our dataset is derived entirely from general regional subreddits (e.g., r/saudiArabia, r/Egypt, r/jordan), from which we extract mental-health-relevant content; this subreddit pool is itself unevenly represented in the resulting corpus rather than uniform across regions (Appendix A). Our extraction pipeline follows the self-reported diagnosis paradigm established by Coppersmith, Dredze, and Harman (2014a) and adopted in subsequent corpus-construction work (Coppersmith et al. 2015; Cohan et al. 2018), which identifies diagnosed users via precision-oriented pattern matching over diagnosis-disclosure phrases rather than clinical verification (See section 3.2 for more discussion about self reported diagnosis). We use a curated list of condition keywords (e.g., اكتئاب حاد, "Severe Depression") and diagnosis phrases (e.g., تم تشخيصي بـ, "I was diagnosed with"), checking whether they co-occur within a 40-character window (measured at the Unicode character level), the same window width used by Cohan et al. (2018) in constructing SMHD, where a sweep over candidate distances found 40 characters to maximize $F_{0.5}$ (precision-weighted) on their validation set. See Appendix C and Appendix D for a detailed pattern description. To enhance inclusivity and robustness, the phrase list incorporates variations across Arabic dialects, including Modern Standard Arabic, Egyptian, and Levantine, along with frequent spellings and misspellings to account for the diverse posting habits of social media users.

3.2 Grounding Self-Disclosure-Based Group Assignment

Both our diagnosed and control groups are assigned entirely from users' voluntary public self-disclosures rather than clinical verification, following the self-reported diagnosis paradigm discussed in §2.1 (Coppersmith, Dredze, and Harman 2014a; Coppersmith et al. 2015; Cohan et al. 2018); De Choudhury et al. (2013) establishes that such self-identified users exhibit measurable linguistic and behavioral differences from the general population, motivating self-disclosure as a scalable diagnosis proxy at corpus scale.

We explicitly acknowledge the limitations this design carries. First, self-disclosure is a proxy for diagnosis, not a diagnosis itself: users may exaggerate or use diagnostic language loosely, genuinely diagnosed users who never post a matching disclosure are missed (false negatives), and sarcastic, fictional, or third-person mentions can pass pattern matching,

²https://github.com/ArthurHeitmann/arctic_shift

³We obtained an exempt IRB approval for this study from our institution, exempt under DHHS regulatory category 4.

Dataset	Condition	Size	Domain	Anno.
MentalQA (Alhuzali, Alasmari, and Alsaleh 2024)	N/A	1,000	Altibbi	Human
OCDTweets (Al-Haider et al. 2024)	OCD symptoms	5K	X	Human
Arabic Suicidal Tweets (Abdulsalam, Alhothali, and Al-Ghamdi 2024)	Suicidal ideation	5.7K	X	Human
Suicide Tweets (Baghdadi et al. 2022)	Suicidal ideation	2.0K	X	Human
AraDepSu (Hassib et al. 2022)	Depression, Suicidal ideation	20K	X	Human
Detecting Depression (Almouzini, Alageel et al. 2019)	Depression	2.7K	X	Human
CairoDep (El-Ramly et al. 2021)	Depression	7K	X	Human
Mood Change (Maghraby and Ali 2022)	Depression	49K	X	Auto.
Ours	12 conditions	625K	Reddit	Auto.

Table 1: Size and coverage comparison of Arabic Mental Health Corpora.

we estimate the resulting irregularity rate through manual annotation (§4.1). Second, control users are only *negatively* defined by the absence of detected mental-health language across their retrieved history, so the control group is better understood as a screened general population than a clinically verified healthy cohort. These limitations are inherent to the self-reported diagnosis paradigm and are shared by the prior work it is based on (Coppersmith, Dredze, and Harman 2014a; Cohan et al. 2018).

3.3 Preprocessing

We apply two filtering steps to the collected post histories. First, we remove all posts containing mental health-related content (self-disclosures, condition keywords, and medication mentions)⁴ to prevent models from relying on explicit diagnostic markers when learning to classify users. This design encourages models to learn more subtle linguistic and behavioral signals rather than trivial keyword-based cues, aligning with prior work in mental health NLP (Cohan et al. 2018). Second, we filter out non-Arabic content using the FastText library to ensure linguistic consistency across the corpus. Any post containing 20% or more non Arabic content was excluded. In addition, we remove posts with fewer than 10 words, including posts marked as [removed] by the Reddit API.⁵

3.4 Control Dataset

We further construct a control dataset of 2,875 users in a similar manner by scraping entire public user post histories from general Arabic-speaking subreddits (e.g., r/SaudiArabia) for candidate control users. Control users are subject to strict exclusion criteria: we immediately discard a candidate if they have ever posted about a mental health-related issue on the platform. For this filtration, we use the same pattern developed to identify diagnosed users, scanning for matching phrases, medication keywords, or other mental health-related terminology. See Appendix D for more detail. Such stringent exclusion criteria are inspired by previous literature (Cohan

et al. 2018) to minimize the risk of including affected users in the control group.

4 Annotation and Validation

4.1 Human Annotation

Two native Arabic-speaking annotators independently labeled a stratified sample of 240 posts from CARMA (20 posts per retained condition) under a binary scheme: TP (a genuine first-person self-disclosure of a mental health diagnosis) or FP (a clear mismatch: sarcasm, third-person reference, figurative use, or negation). Appendix F reproduces the full written guidelines, including worked examples for each label, given to both annotators. The two annotators achieve substantial agreement ($\kappa = 0.701$, 89.8% raw agreement on the $n=235$ posts both annotators labeled⁶), which falls in the “substantial” band (0.61–0.80) on the interpretation scale of Landis and Koch (1977) used throughout this work (Appendix G). Per-label F1 (%) between the two human annotators is 93.5 for TP and 76.5 for FP.

Table 13 reports the full confusion matrix between the two annotators on the 235 jointly-labeled posts, which we use below to derive a corpus-wide estimate of the pipeline’s residual false-positive rate.

Estimated false-positive rate. Ann-1 labeled 57/235 (24.3%) of the sample FP, while Ann-2 labeled 45/235 (19.1%) FP. We treat each annotator’s FP proportion as an independent noisy estimate of the pipeline’s true false-positive rate and take their mean as our point estimate: (21.7%, with the individual annotator rates (19.1–24.3%) indicating the spread of this estimate. We therefore estimate that roughly one in five posts surfaced by the pattern-based extraction pipeline is a false positive (sarcasm, third-person reference, figurative use, or negation), which we report as a caveat on corpus precision (§4.1; Limitations). A residual false-positive rate of this magnitude is consistent with prior manually-validated self-report extraction pipelines for social-media mental health corpora, which likewise rely on human spot-checks to characterise imperfect precision rather than achieving it by construction (Cohan et al. 2018; Coppersmith, Dredze, and Harman 2014b). We describe the pipeline as

⁴We used a list of 305 medications and 140 condition specific terms. Any post that contained an item of this list was excluded from the dataset. 14,621 posts were excluded.

⁵Arctic Shift API allows users to submit requests to remove their posts

⁶Five posts were excluded due to a missing annotation by at least one of the annotators

precision-oriented rather than *high-precision* in an unqualified sense: an estimated 78.3% precision is a substantial improvement over unconstrained condition-keyword matching (which would also match third-person mentions, negations, and sarcasm with no co-occurrence or phrase constraint at all), but it is not the near-perfect precision that phrase “high precision” might otherwise imply, and we flag the residual 21.7% false-positive rate as a corpus-precision caveat throughout (Limitations). We restrict this headline estimate to the two human annotators; §4.2 summarizes a parallel experiment adding two LLM judges as additional raters and why we do not fold them into this estimate, with the full analysis in Appendix G.

4.2 LLM Judges as a Robustness Check

As a robustness check, we additionally had two LLM judges, Qwen2-7B-Instruct (Yang et al. 2024) and Jais-2-8B-Chat (Anwar et al. 2025), independently label the same 240-post sample under the identical TP/FP scheme. Agreement with the human annotators is far weaker than agreement between the two humans ($\kappa=0.138-0.231$ vs. $\kappa=0.701$; Table 14, Appendix G), and both LLMs report a markedly lower FP rate ($\approx 11-14\%$ vs. $\approx 18-23\%$ for the humans), recovering only 21–27% of the posts humans labeled FP. Rather than disagreeing at random, the LLM judges default to TP on the same sarcasm-, negation-, and figurative-language cases that limit this pipeline’s precision, consistent with known LLM limitations on Arabic dialectal sarcasm (Abu Farha and Magdy 2020), negation (García-Ferrero et al. 2023), and leniency bias in LLM-as-judge settings (Jain et al. 2025). We therefore exclude the LLM judges from the headline false-positive-rate estimate of §4.1; full pairwise statistics are in Appendix G.

5 Corpus Statistics

5.1 Overview

The resulting corpus comprises 4,365 unique diagnosed users and their associated 625K non mental-health posts. Table 2 presents the per-condition distribution of users, posts, and post-length statistics for the 12 retained conditions.

Once a user’s self-disclosure post is confirmed by the pattern-based extraction pipeline, we retrieve their complete posting history to enable longitudinal analysis beyond the initial self-disclosure post.

Comorbidity. The pattern-based extraction pipeline matches self-disclosure posts independently per condition, so a user can, prior to filtering, be matched by more than one condition’s phrase-keyword pattern. 394 users (8.3%) have self-disclosure matches for two or more conditions. The most frequent comorbid pairs are depression+anxiety (86 users), depression+ADHD (80), depression+OCD (40), ADHD+anxiety (33), and depression+sleep disorder (32), broadly consistent with known comorbidity patterns in the clinical literature. None of these 394 comorbid users are present in the final retained corpus of Table 2: rather than assigning a comorbid user to a single primary condition, the classification pipeline (§6) excludes comorbid users

Condition	Users	Posts	Mean len.	SD
Depression	1,831	251,809	42.2	63.8
Anxiety	644	90,082	40.4	59.0
ADHD	643	101,107	41.7	59.2
Sleep Disorder	357	60,666	36.5	47.1
OCD	311	39,413	45.3	81.1
Panic Disorder	147	20,849	40.1	52.3
BPD	98	10,735	41.9	55.5
PTSD	92	15,627	42.9	52.3
Suicidal Ideation	80	12,266	46.1	61.3
Bipolar Disorder	76	8,894	38.7	48.9
Schizophrenia	45	7,703	61.5	116.4
Eating Disorder	41	5,386	35.4	44.1
Total	4,365	624,537	41.7	62.1

Table 2: Distribution of users and posts across the 12 retained mental health conditions in CARMA.

entirely, so that each condition’s positive class is free of other-condition self-disclosures. This design isolates condition-specific linguistic signals and avoids label ambiguity, providing a cleaner benchmark for single-condition classification; modeling comorbidity is an important direction for future work.

5.2 Language and Dialect

Dialect identification was performed using CAMEL-BERT-Mix DID NADI Model⁷ (Inoue et al. 2021), a BERT-based transformer fine-tuned on the 21-label NADI country-level dataset for Arabic dialect classification; Inoue et al. (2021) report this model among the strongest of the DID variants they evaluate on the NADI country-level benchmark. We did not separately validate the model’s accuracy on CARMA’s own Reddit text, so some domain shift relative to its original (Twitter-sourced) NADI training/evaluation data is possible; we treat the resulting dialect labels as a description of the corpus rather than a validated ground truth, consistent with how the extraction pipeline’s own precision is treated (§4.1). As shown in Figure 1, Egyptian Arabic is the dominant dialect in CARMA (56.1%), with Egypt as the primary contributing country, consistent with its status as one of the most populous Arabic-speaking countries with approximately 116 million inhabitants as of 2024⁸ (United Nations, Department of Economic and Social Affairs, Population Division 2024). Gulf Arabic constitutes the second-largest group, primarily driven by Saudi Arabia (approximately 35 million inhabitants). Levantine and Maghrebi Arabic together account for under 7% of the corpus. This distribution reflects the strong presence of Egyptian Arabic speakers on Reddit and their engagement in public regional subreddits, and is corroborated directly by the subreddit-level provenance of the corpus’s self-disclosure posts (Appendix A, Figure 2): although the extraction pipeline’s phrase and keyword patterns are de-

⁷<https://huggingface.co/CAMEL-Lab/bert-base-arabic-camelbert-mix-did-nadi>

⁸Population estimates sourced from official 2024 UN demographic data.

signed to support Gulf, Egyptian, Levantine, and Modern Standard Arabic phrasing (§3), the resulting corpus is not evenly distributed across these dialects, and should not be read as a general or balanced sample of Arabic Reddit.

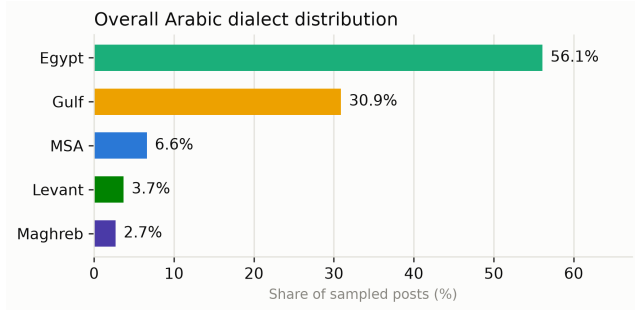


Figure 1: Arabic Dialect Distribution in CARMA

5.3 Lexical-Semantic Differences

To complement the dialect distribution above, we characterize lexical differences between diagnosed and control users using a log-odds-ratio with an informative Dirichlet prior (Monroe, Colaresi, and Quinn 2008) over unigrams (pooled diagnosed vs. pooled control corpus, combined word counts as background prior), which surfaces words distinctively over-used by one group while correcting for raw frequency (a z-score, not a raw rate difference). To keep the ranked vocabulary content-bearing and free of single-user artifacts, we (i) exclude a list of 2,001 stopwords, (ii) require a token to occur at least 10 times in the combined corpus, and (iii) require that, in whichever group a token has nonzero count, at least $k=5$ distinct users contribute to that count. Without this last adjustment, a handful of repetitive or high-volume posters can register as a spurious group-level “keyword”, e.g., one control user alone accounted for all 190 occurrences of a proper noun that otherwise ranked among the top control-associated terms.

Table 3 lists the most distinctive terms per group after these adjustments. Diagnosed-associated terms still include Gulf-Arabic discourse markers and free pronouns/particles (e.g., احس “I feel,” مدري “I don’t know,” ربي “my Lord”), each used by hundreds to thousands of distinct diagnosed users, alongside content words such as الصدق “the truth/honesty” and اشياء “things.” Control-associated terms skew toward proper nouns and topical terms (e.g., الجيش “the army,” سوريا “Syria,” منسف “Mansaf,” a regional dish), consistent with the control corpus drawing more heavily from news/political and regional-lifestyle subreddits. We report this ranking as a description of the corpus rather than as validated clinical signal, since it appears to reflect dialect and topic composition at least as much as condition-specific language. This is not only a caveat on the log-odds ranking itself: the same dialect and topic composition differences plausibly confound the diagnosed-vs-control classification results reported in §7, which we discuss further in Limitations.

Term	Gloss	z	n_D	n_C
<i>Diagnosed-associated</i>				
احس	I feel	15.11	2076	542
مدري	I don’t know (Gulf)	10.58	1062	220
ربي	my Lord	10.43	1213	282
يارب	oh God	9.99	1446	431
دائم	always	9.61	872	191
تي	you want (Gulf)	8.04	846	173
ادري	I know (Gulf)	7.87	1281	346
الصدق	the truth / honesty	7.81	649	141
<i>Control-associated</i>				
الجيش	the army	−10.56	788	459
المجاز	metaphor	−10.54	15	9
متفرقه	miscellaneous	−9.44	63	24
بجامعه	at a university	−9.43	166	55
سوريا	Syria	−8.81	526	306
الهدم	demolition	−8.46	19	7
منسف	Mansaf (regional dish)	−8.45	35	20
تريد	want (MSA)	−8.40	212	119

Table 3: Most distinctive unigrams per group by log-odds-ratio with Dirichlet prior.

6 Experiments

We conducted two binary classification experiments to evaluate the dataset’s utility: classical models trained on TF-IDF bag-of-words features, and pre-trained language models (PLMs) fine-tuned on user text. Both pipelines are evaluated at the *post level* (each post an independent instance, label inherited from its author) and *user level* (a user’s posts aggregated into a single TF-IDF document, or a single truncated token sequence for PLMs), letting us compare how classical and PLM-based classifiers respond to the same shift in input granularity. User-level aggregation trades a much larger number of noisier training instances for fewer, richer per-user representations, but also shrinks the effective test-set size for lower-prevalence conditions (Appendix Tables 5 and 7). Both experiments use a balanced 1:1 ratio of diagnosed to control samples.

6.1 Experimental Setup

We frame the task as binary classification: for each condition C , classify whether a user is diagnosed with C (positive) or is a control user (negative), randomly sampled from Arabic-speaking Reddit users with no self-reported mental health conditions. We split the data into 90% training and 10% test sets, ensuring no user appears across multiple splits; for post-level experiments, posts are grouped by user prior to splitting to prevent leakage.

6.2 Models

Classical Classifiers. We train four traditional machine learning models on TF-IDF bag-of-words features: Naive Bayes (NB), Logistic Regression (LR), Support Vector Machines (SVM), and XGBoost (XGB). These serve as baselines to assess whether simple lexical features can capture condition-specific language patterns.

Pre-trained Language Models (PLMs). We fine-tune two Arabic PLMs:

- **AraBERT-Twitter** (aubmindlab/bert-base-arabertv02-twitter): A BERT model pre-trained on Arabic tweets, well-suited for social media text.
- **CAMELBERT** (CAMEL-Lab/bert-base-arabic-camelbert-mix): A BERT model pre-trained on a diverse mix of Arabic text including Modern Standard Arabic and dialectal content.

For fine-tuning, we use a learning rate of $2e-5$, batch size of 16, and train for 3 epochs with early stopping based on validation F1.

6.3 Evaluation

We evaluate all models using four standard metrics: F1-score, Accuracy, Precision, and Recall. F1-score serves as our primary metric as it balances precision and recall, which is important given the class imbalance across conditions. All metrics are computed on the held-out test set.

7 Results and Discussion

In this work, we introduce a large-scale Arabic mental health corpus comprising over 625K posts spanning 12 diagnosed mental health conditions. The primary objective of this resource is to facilitate future research in Arabic mental health analysis by providing a publicly available benchmark, rather than to establish new state-of-the-art performance on mental health classification.

Table 4 summarizes the best F1 score achieved for each condition by the two modeling pipelines at both the post and user levels (reported as *post level* | *user level*). Complete per-model results are provided in Appendix Tables 5 and 7.

Four considerations are important when interpreting these results. First, the reported scores should be viewed primarily as strong baseline benchmarks for future work rather than as an exhaustive evaluation of the full range of possible modeling approaches. Second, performance on conditions with relatively few diagnosed users (e.g., eating disorders, schizophrenia, bipolar disorder, and suicidal ideation) should be interpreted with caution, as the corresponding test sets are small and therefore produce less stable evaluation estimates. Third, and most importantly, dialectal and topical differences between the diagnosed and control subreddit pools are a plausible confound that neither pipeline can separate from condition-specific signal at the representation level, so the F1 scores below should be read as evidence of separability under the current corpus construction, not as validated clinical signal; see §5.3 and Limitations for the supporting evidence and full discussion. Fourth, the notably high recall scores at the post level (Appendix Table 5) partly reflect a post-count imbalance introduced by our user-level balancing: diagnosed and control users are sampled 1:1 by *user count* (§6.1), but diagnosed users contribute roughly $1.7\text{--}2.6\times$ as many posts per user on average as control users across conditions, so post-level train/test sets skew toward the positive class even though user counts are balanced, biasing classifiers toward predicting positive and inflating recall (at some

Condition	Classical	PLM
Depression	.81 .68	.80 .71
Anxiety	.83 .85	.80 .72
ADHD	.75 .72	.75 .71
Sleep dis.	.81 .89	.78 .73
OCD	.80 .82	.71 .72
Panic	.82 .91	.85 .77
BPD	.86 .72	.85 .71
PTSD	.67 .70	.70 .64
Suicidal	.86 .82	.87 .78
Bipolar	.88 .59	.86 .71
Schizophrenia	.96 .91	.88 .80
Eating dis.	.79 .73	.75 .73

Table 4: Best F1 score per condition on CARMA (Reddit). Each cell reports *post level* | *user level*. Full results in Appendix Tables 5 and 7.

cost to precision) relative to what a post-count-balanced split would show.

7.1 Classical Classifier Performance

Schizophrenia achieves the highest post-level F1 (.956, NB); PTSD is hardest (.665). XGBoost attains the highest post-level mean F1 (.815), narrowly ahead of NB (.812), LR (.798), and SVM (.764). Aggregating to the user level reshuffles both scores and model ranking: SVM becomes strongest on average (.730) despite being weakest at the post level, while NB drops from strongest to weakest (.644), plausibly because NB’s frequent-lexical-marker strength dilutes once posts are merged, whereas SVM’s margin-based boundary is more robust to the resulting feature sparsity. User-level test sets are small for several conditions (as few as 10 users; Appendix Table 5), so individual values should be read with this variance in mind.

7.2 Fine-tuned PLM Performance

AraBERT-Twitter and CAMELBERT perform comparably at both the post level (mean F1 .792 vs. .788) and user level (.702 vs. .700), with neither architecture consistently dominating (AraBERT ahead on 7/12 conditions post-level, 5/12 user-level). Both models drop substantially from post to user level, reflecting information loss from truncating a user’s full history to a single fixed-length input. Schizophrenia is easiest (.875 post, .800 user) and PTSD hardest (.677–.701 post, .588–.640 user) for both models.

7.3 Post-level vs. User-level Aggregation

Aggregation reduced mean F1 for every model (classical: NB $-.168$, LR $-.115$, SVM $-.034$, XGB $-.127$; PLMs: AraBERT $-.090$, CAMELBERT $-.088$), reflecting harder user-level representation (vocabulary dilution, input truncation) and smaller test sets for low-prevalence conditions. SVM was the most robust model to this shift, NB the most brittle. Five classical-pipeline conditions (anxiety, sleep disorder, OCD, panic, PTSD) improved at the user level, versus only a marginal OCD improvement for PLMs, suggesting TF

IDF benefits more from aggregated histories than fine-tuned PLMs.

8 Conclusion

We introduced CARMA, a large-scale Arabic mental health corpus comprising 4,365 diagnosed users across 12 retained mental health conditions and 2,875 control users from Reddit. Our work addresses a critical gap in Arabic NLP resources for mental health research. We developed a dialect-aware, pattern-based extraction pipeline designed to support self-reported diagnoses across Gulf, Egyptian, Levantine, and Modern Standard Arabic, though the resulting corpus is dominated by Egyptian and Gulf Arabic rather than evenly distributed across dialects (§5.2); it was validated through manual annotation by two native Arabic-speaking annotators achieving substantial inter-annotator agreement.

Our baseline experiments, run at both the post level and the user level, demonstrate that both classical classifiers and fine-tuned pre-trained language models can distinguish diagnosed users from controls across nearly all 12 conditions (F1 up to 0.96, schizophrenia), though this separability is confounded with dialect and topic composition (§7; Limitations) and should not be read as validated clinical signal. Both pipelines lose performance on average moving from post-level to user-level aggregation, with classical models more sensitive to this shift than the PLMs.

We release this corpus to facilitate future research in Arabic computational psychiatry, dialectal NLP, and cross-cultural mental health analysis. We hope CARMA will serve as a foundation for developing more inclusive mental health technologies for the Arabic-speaking world.

Limitations

Dialectal and Topical Confounding in Classification Results. A key concern with the classification results of §7 is that reported performance may partially reflect dialectal and topical differences between the diagnosed and control groups rather than condition-specific language. The log-odds analysis of §5.3 indicates that many of the most discriminative features are regional discourse markers (e.g., Gulf-Arabic pronouns and particles) or topic-specific vocabulary (e.g., news/political terms, regional dishes) rather than clinical language, and this pattern is consistent with the regional and dialectal skew documented in §5.2 and Appendix A: diagnosed and control users are drawn from overlapping but not identical pools of subreddits, so dialect and topic composition can differ systematically between the two groups by construction, independent of the target condition. Neither the TF-IDF nor the fine-tuned PLM pipeline is able to separate condition-specific signal from these confounds at the representation level, and we did not run experiments that explicitly control for dialect or topic (e.g., dialect-matched control sampling, or ablations that remove the log-odds-flagged terms). Frequency- or subreddit-matched control sampling, as used in constructing SMHD (Cohan et al. 2018) and MINDSET (Mankarious et al. 2025), was feasible in those corpora because English-language Reddit offers a large enough pool of candidate control users and matched subreddits to sample

from; the Arabic-speaking subreddit ecosystem is comparatively small (§3, Appendix A), so restricting control sampling to subreddits or posting-frequency bands that match the diagnosed group would substantially shrink an already modest control pool and is not currently feasible at this corpus’s scale. We therefore cannot currently quantify how much of the reported classification performance is attributable to genuine condition-specific signal versus dialectal or topical separability, and recommend the classification results in §7 be interpreted as evidence of separability between diagnosed and control text under the current corpus construction, rather than as validated clinical signal, until controlled analyses of this kind are conducted. Designing and validating such controls, e.g., dialect-matched control sampling, or topic-adjusted feature ablations, is a substantial undertaking in its own right and is out of scope for the present work, whose primary contribution is the construction, validation, and release of the CARMA corpus itself rather than a definitive classification benchmark; we leave this confound analysis to future work built on the released resource.

Annotation Pipeline Reliability. The pattern-based extraction pipeline prioritises precision through strict phrase-keyword co-occurrence matching (Appendix C, Appendix D), which may miss self-disclosures phrased outside the curated phrase and keyword lists, limiting recall. Our human validation on a 240-post sample indicates substantial but imperfect precision (Ann-1 vs. Ann-2 $\kappa = 0.701$, 89.8% agreement; §4.1), from which we estimate a residual false-positive rate of approximately 21.7% (mean of the two annotators’ individual FP rates, 19.1–24.3%; §4.1) in the corpus.

Temporal Validity of Posting History. Once a user’s disclosure post is confirmed, we retrieve and use their *entire* available posting history (§5.1) without regard to when each post falls relative to the disclosure event: a post written years before the disclosure – potentially before the condition’s onset or before the user’s own diagnosis – is treated identically to one written immediately after it. Because the released per-post fields do not retain a timestamp for every post in a user’s complete history (only the confirming self-disclosure posts retain `created_utc`; Appendix A), we cannot directly quantify what fraction of each user’s retained posts precede versus follow their disclosure. This is a limitation inherited from the underlying self-reported-diagnosis paradigm (§3.2) rather than one specific to CARMA, but it means linguistic patterns attributed to a condition may in part reflect a user’s language at any point in their posting history, not specifically the period during or after diagnosis.

Input Truncation in PLM Baselines. The user-level PLM experiments (§6.1) concatenate a user’s posts into a single sequence truncated to the model’s maximum input length, discarding whatever text falls beyond that limit and collapsing the temporal order of a user’s posting history into one static representation. This is a likely contributor to the post-to-user performance drop observed for both AraBERT-Twitter and CAMELBERT (§7) and means the user-level PLM results should be read as a baseline for a single-sequence encoding

of user history, not as an upper bound on what a history-aware architecture (e.g., hierarchical or windowed encoding of a user’s posts) could achieve on this corpus.

Regional and Dialectal Skew. Although the extraction pipeline’s phrase and keyword patterns are designed to support Gulf, Egyptian, Levantine, and Modern Standard Arabic, the resulting corpus is not a balanced sample across these varieties: Egyptian and Gulf Arabic together dominate, while Levantine and Maghrebi Arabic combined account for under 7% of the corpus, driven by a small number of Egyptian- and Saudi-focused subreddits contributing the majority of confirmed self-disclosure posts (§5.2, Appendix A). Posting volume is also heavily concentrated in recent years (2024 onward; Appendix A, Figure 3). Findings and benchmark results should therefore not be read as representative of Arabic Reddit, or of Arabic-speaking communities, in general.

Sensitive Data Handling. This work involves sensitive personal health information derived from social media; see Ethical Considerations below for how we mitigate re-identification and misuse risks.

Ethical Considerations

This study involves publicly available social media data from users who have voluntarily disclosed their mental health status. We acknowledge the sensitive nature of mental health data and received IRB approval from our institution in regards to the collection and distribution of this data.

All data collection adheres to Reddit’s Terms of Service and API usage guidelines at the time of collection. We use pre-downloaded public archives from Arctic Shift, which contain posts from publicly accessible Arabic-speaking subreddits, collecting only content that users have chosen to share publicly.

Despite our best efforts to anonymize data, residual re-identification risks may remain. The released corpus consists of post-level text and per-user groupings with Reddit usernames replaced by irreversibly hashed pseudonymous identifiers, with all self-disclosure posts removed from each user’s released history (§8). Consistent with best practices in social media mental health research (Harrigian, Aguirre, and Dredze 2021), the dataset is distributed exclusively under a Data Use Agreement (DUA) that restricts access to qualified researchers, prohibits attempts to re-identify individuals or re-link hashed identifiers to Reddit accounts, and prohibits redistribution.

The corpus is intended solely for research purposes, specifically to advance Arabic NLP for mental health detection. We recognize that users who disclose mental health conditions on social media may not anticipate their posts being used in research. We mitigate this concern by removing all self-disclosure posts from users’ released posting histories, ensuring that the data cannot be used to re-identify individuals based on their diagnostic disclosures. Researchers using this corpus should adhere to ethical guidelines for mental health research and avoid applications that could stigmatize or harm the individuals represented in the data.

Dual-Use Risk. A resource of this kind carries a dual-use risk beyond the individual re-identification risks addressed above: the same detection methodology could in principle be repurposed for non-consensual, individual-level mental-health profiling or screening at scale (e.g., by employers, insurers, educational institutions, or state actors) rather than for the aggregate, corpus-level research CARMA is intended for. We consider this risk to be elevated, not merely generic, in the specific social and legal context this corpus covers: across several of the jurisdictions the corpus’s subreddits are drawn from, disclosed mental illness can carry significant social stigma and, in some cases, legal or occupational consequences (e.g., around employment, licensing, or family law), consequences that a self-disclosed, non-clinically-verified label (§3.2) makes especially inappropriate to act on for any individual. We do not release any tooling packaged for individual-level screening, gate the corpus behind a DUA that restricts use to qualified researchers and prohibits re-identification (above), and explicitly discourage any deployment of models trained on this corpus for individual diagnostic, screening, employment, insurance, or law-enforcement use; the corpus and any derived models are intended for aggregate, population- and dataset-level research only.

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9 Paper Checklist

1. For most authors...

- (a) Would answering this research question advance science without violating social contracts, such as violating privacy norms, perpetuating unfair profiling, exacerbating the socio-economic divide, or implying disrespect to societies or cultures? **Yes, and: this work fills a documented gap in Arabic-language mental-health NLP resources (§1), and we engage directly with the social-contract risks it raises – misleading terminology, dual-use, and stigma – in Ethical Considerations.**
- (b) Do your main claims in the abstract and introduction accurately reflect the paper’s contributions and scope? **Yes, see the Abstract and the contributions list in §1.**
- (c) Do you clarify how the proposed methodological approach is appropriate for the claims made? **Yes, see §3–4, where the extraction pipeline is described and its precision is empirically validated via manual annotation rather than assumed.**
- (d) Do you clarify what are possible artifacts in the data used, given population-specific distributions? **Yes, see §5.2 and the Limitations section (Dialectal and Topical Confounding; Regional and Dialectal Skew).**
- (e) Did you describe the limitations of your work? **Yes, see Limitations.**
- (f) Did you discuss any potential negative societal impacts of your work? **Yes, see Ethical Considerations, including the Dual-Use Risk paragraph.**
- (g) Did you discuss any potential misuse of your work? **Yes, see the Dual-Use Risk paragraph in Ethical Considerations.**
- (h) Did you describe steps taken to prevent or mitigate potential negative outcomes of the research, such as data and model documentation, data anonymization, responsible release, access control, and the reproducibility of findings? **Yes, see Ethical Considerations: IRB approval, hashed pseudonymous identifiers, removal of self-disclosure posts, and Data Use Agreement (DUA) gated release.**
- (i) Have you read the ethics review guidelines and ensured that your paper conforms to them? **Yes.**

2. Additionally, if your study involves hypotheses testing...

- (a) Did you clearly state the assumptions underlying all theoretical results? **NA: this paper does not test formal social-science hypotheses; it introduces a corpus and baseline classification results.**
- (b) Have you provided justifications for all theoretical results? **NA.**
- (c) Did you discuss competing hypotheses or theories that might challenge or complement your theoretical results? **NA.**
- (d) Have you considered alternative mechanisms or explanations that might account for the same outcomes observed in your study? **NA.**
- (e) Did you address potential biases or limitations in your theoretical framework? **NA.**

- (f) Have you related your theoretical results to the existing literature in social science? **NA.**
- (g) Did you discuss the implications of your theoretical results for policy, practice, or further research in the social science domain? **NA.**

3. Additionally, if you are including theoretical proofs...

- (a) Did you state the full set of assumptions of all theoretical results? **NA: this paper contains no theoretical proofs.**
- (b) Did you include complete proofs of all theoretical results? **NA.**

4. Additionally, if you ran machine learning experiments...

- (a) Did you include the code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL)? **Yes, and: the corpus is released under a Data Use Agreement restricting access to qualified researchers (Ethical Considerations); training and evaluation instructions are given in §6.1–6.2.**
- (b) Did you specify all the training details (e.g., data splits, hyperparameters, how they were chosen)? **Yes, see §6.1 (90/10 split, user-level grouping to prevent leakage) and §6.2 (learning rate, batch size, epochs, early stopping).**
- (c) Did you report error bars (e.g., with respect to the random seed after running experiments multiple times)? **No, results are reported for a single run rather than averaged across seeds; we do note where small test-set sizes make individual conditions’ estimates higher-variance (§7, Appendix Tables 5–7).**
- (d) Did you include the total amount of compute and the type of resources used (e.g., type of GPUs, internal cluster, or cloud provider)? **No, compute resources are not itemized in this submission.**
- (e) Do you justify how the proposed evaluation is sufficient and appropriate to the claims made? **Yes, see §6.3 for the choice of metrics, and §7 for the interpretive caveats (small test sets, dialectal/topical confounding, post-level class skew) that scope what the results do and do not support.**
- (f) Do you discuss what is “the cost” of misclassification and fault (in)tolerance? **No formal cost analysis is included; the Dual-Use Risk paragraph in Ethical Considerations addresses related harms from acting on model outputs at the individual level, but does not quantify misclassification cost.**

5. Additionally, if you are using existing assets (e.g., code, data, models) or curating/releasing new assets, **without compromising anonymity**...

- (a) If your work uses existing assets, did you cite the creators? **Yes, e.g. Arctic Shift (§3), AraBERT-Twitter and CAMELBERT (§6.2), Qwen2-7B-Instruct and Jais-2-8B-Chat (§4.2), and the CAMELBERT-Mix DID NADI dialect model (§5.2).**

- (b) Did you mention the license of the assets? **No, the licenses of the pretrained models and archived data source used are not enumerated in this submission.**
- (c) Did you include any new assets in the supplemental material or as a URL? **Yes, and: the CARMA corpus itself is the new asset, released under a Data Use Agreement (Ethical Considerations).**
- (d) Did you discuss whether and how consent was obtained from people whose data you're using/curating? **Yes, and: no individual consent was obtained given the public nature of the data; this is addressed via IRB exemption (DHHS category 4) and mitigated through DUA-gated release and removal of self-disclosure posts (Ethical Considerations).**
- (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **Yes, see Ethical Considerations (sensitive mental-health information, re-identification risk) and §8.**
- (f) If you are curating or releasing new datasets, did you discuss how you intend to make your datasets FAIR (see Wilkinson et al. (2016))? **No, the FAIR framework is not discussed explicitly, though the DUA-gated release (Ethical Considerations) addresses accessibility to qualified researchers.**
- (g) If you are curating or releasing new datasets, did you create a Datasheet for the Dataset (see Gebru et al. (2021))? **No, a formal Datasheet is not included in this submission.**
6. Additionally, if you used crowdsourcing or conducted research with human subjects, **without compromising anonymity...**
- (a) Did you include the full text of instructions given to participants and screenshots? **Yes, and: Appendix F reproduces the full annotation guidelines given to annotators; no screenshots are included, as annotation was not conducted through a crowdsourcing platform interface.**
- (b) Did you describe any potential participant risks, with mentions of Institutional Review Board (IRB) approvals? **Yes, see Ethical Considerations (exempt IRB approval, DHHS regulatory category 4).**
- (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **No, annotator compensation is not described in this submission.**
- (d) Did you discuss how data is stored, shared, and de-identified? **Yes, see Ethical Considerations (hashed pseudonymous identifiers, removal of self-disclosure posts, DUA-gated distribution).**

A Subreddit and Temporal Distribution

The released post data does not retain a subreddit or timestamp field for every post in a user's complete retrieved history, so a per-post breakdown over the full ~625K-post corpus is not available. We can, however, characterize the provenance of the posts that *seeded* each user's inclusion

in CARMA: the self-disclosure posts confirmed by the pattern-based extraction pipeline (§3) do retain subreddit and created_utc metadata. Figure 2 plots the subreddit distribution of these 6,457 confirmed self-disclosure posts (across the 12 retained conditions), and Figure 3 plots their monthly volume over time.

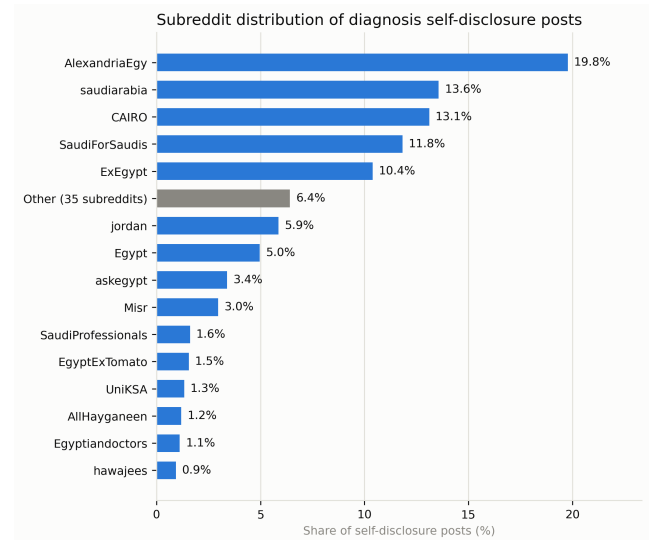


Figure 2: Subreddit distribution of the self-disclosure posts confirming diagnosed users in CARMA. The top 5 subreddits account for roughly 69% of confirmed disclosures; the remaining 35 subreddits contribute the rest.

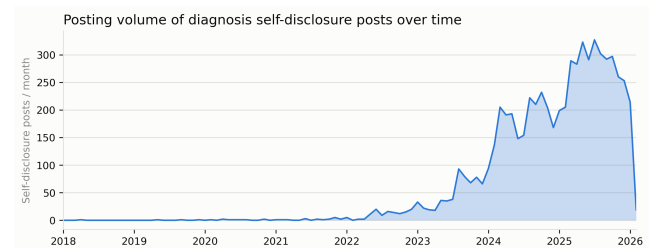


Figure 3: Monthly volume of self-disclosure posts confirming diagnosed users in CARMA, 2018–2026. Volume is negligible before 2022 and rises sharply from 2024 onward.

Two observations follow directly from these figures and are referenced in §3 and §5.2. First, subreddit coverage is highly concentrated: the five largest subreddits (predominantly Egyptian- and Saudi-focused: *r/AlexandriaEgy*, *r/saudiarabia*, *r/CAIRO*, *r/SaudiForSaudis*, *r/ExEgypt*) account for roughly 69% of confirmed self-disclosure posts, while Levantine-associated subreddits (e.g., *r/jordan*) contribute under 6% and no clearly Maghrebi-associated subreddit appears among the top contributors. This subreddit concentration is consistent with, and likely a direct driver of, the dialect skew reported in §5.2 (Egyptian and Gulf Arabic dominant; Levantine and Maghrebi Arabic together under 7%). Second, posting volume is heavily skewed to-

B Full Classification Results

Tables 5–6 (classical classifiers, split across two tables for length) and 7 (PLMs) show the entire results from our two classification experiments.

C Condition Keyword Lists

D Self-Disclosure Phrase Inventory

Tables 9 and 10 list the full self-disclosure phrase inventory (117 universal + 30 dialect-specific not in the universal set = 147 total unique phrases). Universal phrases (Table 9) apply across all dialects; they are organized by semantic type and connector class. Dialect-specific phrases (Table 10) supplement the universal set with register-appropriate variants for Gulf, Levantine, Egyptian, Maghrebi, and Modern Standard Arabic (MSA). Orthographic variants (hamza / hamza-less; taa marbuta ﺀ / ﻩ) are generated automatically at runtime and not listed separately.

Table 11 summarizes the three connector types used by the pipeline, showing how each phrase class links to the condition keyword and how the definite article *ال* triggers the *بال* fusion rule.

The pipeline generates all orthographic variants automatically. Table 12 lists the variant pairs handled.

Both human annotators received the following written guidelines (translated from Arabic and lightly edited for clarity, but otherwise reproduced in full) prior to labeling. Annotators judged each post together with its pipeline-assigned *diagnosis* column and assigned exactly one of two labels: TP or FP.

"I have depression." , انا عندي اكتئاب
"I was diagnosed with depression." , اتشخصت بـ اكتئاب

“My brother has depression.” (third-person reference)
 “Depression is going to kill me, hahaha.” (sarcasm)

Cohen's κ values are interpreted following Landis and Koch (1977): $\kappa \leq 0.20$ slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, 0.81–1.00 almost perfect agreement.

We do not fold the LLM judges into the headline estimate, because the gap above reflects a systematic bias rather than additional independent noise around the same true rate. Conditioning on the posts each human annotator labeled FP, the LLM judges recover only a minority of them: of Ann-1’s 51 (vs. Qwen) and 52 (vs. Jais) FP calls, Qwen agrees on 21.6% and Jais on 25.0%; of Ann-2’s 38 (vs. Qwen) and 41 (vs. Jais) FP calls, Qwen agrees on 21.1% and Jais on 26.8%. In other words, LLM judges default to TP on many posts that both human annotators independently flag as sarcasm, third-person reference, figurative use, or negation – the same non-literal constructions the pattern-based pipeline itself struggles with. Their lower apparent FP rate is therefore evidence of lower sensitivity to these cases, not evidence that the corpus is cleaner than the human-only estimate suggests.

This pattern is consistent with three independent strands of prior work. First, sarcasm and other forms of dialectal figurative language are a documented weak point for Arabic NLP specifically, motivating dedicated resources such as the Ar-Sarcasm dataset (Abu Farha and Magdy 2020). Second, negation handling is a well-known blind spot for large language models more broadly, which tend to default to the affirmative reading of a statement even when a negation cue is present (García-Ferrero et al. 2023). Third, using LLMs as binary judges has independently been shown to induce a general *leniency* (or “agreeableness”) bias – high agreement on positive cases alongside a much lower rate of correctly identifying negative ones (Jain et al. 2025), mirroring the asym-

Table 5: Full classical classifier results on CARMA (Reddit), conditions 1–6 of 12 (continued in Table 6). Each cell reports *post level* | *user level*: post-level instances treat each post as an independent, author-labeled example, while user-level instances aggregate a user’s posts into a single TF-IDF document. User-level test sets are small for rare conditions; these F1 values are correspondingly higher-variance than the post-level results.

Condition	Model	F1	Acc.	P	R
Depression	NB	0.807 0.488	0.679 0.606	0.681 0.697	0.992 0.375
	LR	0.791 0.676	0.667 0.677	0.690 0.678	0.926 0.674
	SVM	0.787 0.674	0.664 0.663	0.690 0.653	0.917 0.696
	XGB	0.808 0.683	0.680 0.682	0.681 0.681	0.994 0.685
Anxiety	NB	0.819 0.661	0.696 0.669	0.714 0.677	0.960 0.646
	LR	0.798 0.744	0.676 0.761	0.721 0.804	0.894 0.692
	SVM	0.782 0.797	0.659 0.808	0.721 0.845	0.853 0.754
	XGB	0.828 0.850	0.709 0.862	0.717 0.927	0.979 0.785
ADHD	NB	0.738 0.576	0.590 0.615	0.602 0.641	0.951 0.523
	LR	0.733 0.667	0.592 0.654	0.608 0.643	0.921 0.692
	SVM	0.724 0.662	0.590 0.646	0.612 0.634	0.885 0.692
	XGB	0.751 0.721	0.603 0.685	0.607 0.646	0.984 0.815
Sleep dis.	NB	0.810 0.765	0.684 0.778	0.686 0.812	0.988 0.722
	LR	0.781 0.754	0.655 0.764	0.688 0.788	0.903 0.722
	SVM	0.752 0.771	0.629 0.778	0.690 0.794	0.827 0.750
	XGB	0.802 0.886	0.671 0.889	0.680 0.912	0.976 0.861
OCD	NB	0.746 0.594	0.612 0.594	0.716 0.594	0.778 0.594
	LR	0.716 0.739	0.597 0.734	0.740 0.727	0.692 0.750
	SVM	0.683 0.742	0.570 0.750	0.742 0.767	0.633 0.719
	XGB	0.797 0.821	0.672 0.844	0.728 0.958	0.882 0.719
Panic	NB	0.825 0.722	0.734 0.667	0.767 0.619	0.892 0.867
	LR	0.824 0.812	0.727 0.800	0.754 0.765	0.909 0.867
	SVM	0.804 0.909	0.711 0.900	0.770 0.833	0.841 1.000
	XGB	0.818 0.788	0.717 0.767	0.747 0.722	0.903 0.867

metry we observe here (Qwen and Jais both agree with humans far more often on TP than on FP; Table 14). Together these strands suggest our LLM judges’ lower FP rate reflects known, generic limitations of LLM-based annotation rather than a sample-specific artifact.

Illustrative Disagreement Examples

Table 15 illustrates the four most common disagreement patterns among the four raters (cf. the disagreement breakdown underlying Table 14) with paraphrased examples. To protect user privacy, consistent with our policy of removing self-disclosure posts from the released corpus (§8), these examples are *paraphrased reconstructions* illustrating the phenomenon behind each disagreement pattern rather than verbatim quotations from CARMA.

Table 6: Full classical classifier results on CARMA (Reddit), conditions 7–12 of 12 (continued from Table 5). Each cell reports *post level* | *user level*. User-level test sets are small for rare conditions (e.g., $n=10$ for eating disorder and schizophrenia, $n=16$ for bipolar disorder and suicidal ideation); these F1 values are correspondingly higher-variance than the post-level results.

Condition	Model	F1	Acc.	P	R
BPD	NB	0.863 0.720	0.778 0.650	0.808 0.600	0.927 0.900
	LR	0.825 0.556	0.729 0.600	0.804 0.625	0.849 0.500
	SVM	0.765 0.600	0.661 0.600	0.802 0.600	0.731 0.600
	XGB	0.858 0.571	0.769 0.550	0.799 0.545	0.927 0.600
PTSD	NB	0.665 0.700	0.550 0.700	0.514 0.700	0.944 0.700
	LR	0.665 0.444	0.567 0.500	0.525 0.500	0.906 0.400
	SVM	0.632 0.667	0.559 0.650	0.523 0.636	0.798 0.700
	XGB	0.655 0.500	0.539 0.500	0.507 0.500	0.924 0.500
Suicidal	NB	0.861 0.667	0.772 0.562	0.786 0.538	0.953 0.875
	LR	0.855 0.706	0.764 0.688	0.786 0.667	0.936 0.750
	SVM	0.813 0.824	0.715 0.812	0.792 0.778	0.835 0.875
	XGB	0.864 0.750	0.776 0.750	0.784 0.750	0.963 0.750
Bipolar	NB	0.883 0.571	0.791 0.438	0.792 0.462	0.998 0.750
	LR	0.874 0.526	0.778 0.438	0.793 0.455	0.975 0.625
	SVM	0.803 0.556	0.682 0.500	0.790 0.500	0.816 0.625
	XGB	0.871 0.588	0.772 0.562	0.789 0.556	0.971 0.625
Schizophrenia	NB	0.956 0.769	0.915 0.700	0.917 0.625	0.998 1.000
	LR	0.949 0.909	0.903 0.900	0.922 0.833	0.977 1.000
	SVM	0.875 0.833	0.782 0.800	0.919 0.714	0.835 1.000
	XGB	0.943 0.600	0.892 0.600	0.915 0.600	0.972 0.600
Eating dis.	NB	0.773 0.500	0.651 0.400	0.801 0.429	0.746 0.600
	LR	0.761 0.667	0.642 0.600	0.811 0.571	0.716 0.800
	SVM	0.745 0.727	0.628 0.700	0.819 0.667	0.683 0.800
	XGB	0.789 0.500	0.667 0.400	0.795 0.429	0.783 0.600

Table 7: Full fine-tuned PLM results on CARMA (Reddit). Each cell reports *post level* | *user level*: post-level instances classify each post individually (label inherited from its author), while user-level instances concatenate a user’s posts into a single (truncated) input sequence. Best F1 per condition in **bold** for each level. User-level test sets are small for rare conditions (e.g., $n=10$ for eating disorder and schizophrenia, $n=16$ for bipolar disorder and suicidal ideation); these F1 values are correspondingly higher-variance than the post-level results.

Condition	Model	F1	Acc.	P	R
Depression	AraBERT	0.796 0.707	0.680 0.674	0.702 0.642	0.918 0.788
	CAMeLBERT	0.789 0.606	0.670 0.647	0.697 0.685	0.908 0.543
Anxiety	AraBERT	0.766 0.696	0.643 0.623	0.722 0.583	0.817 0.862
	CAMeLBERT	0.800 0.719	0.678 0.700	0.720 0.676	0.901 0.769
ADHD	AraBERT	0.747 0.672	0.622 0.661	0.629 0.652	0.920 0.692
	CAMeLBERT	0.749 0.710	0.629 0.692	0.635 0.671	0.914 0.754
Sleep dis.	AraBERT	0.777 0.719	0.655 0.653	0.695 0.604	0.881 0.889
	CAMeLBERT	0.773 0.727	0.652 0.708	0.697 0.683	0.867 0.778
OCD	AraBERT	0.709 0.571	0.605 0.625	0.771 0.667	0.656 0.500
	CAMeLBERT	0.711 0.721	0.594 0.625	0.744 0.574	0.680 0.969
Panic	AraBERT	0.851 0.750	0.777 0.733	0.803 0.706	0.906 0.800
	CAMeLBERT	0.848 0.774	0.773 0.767	0.802 0.750	0.898 0.800
BPD	AraBERT	0.847 0.696	0.766 0.650	0.836 0.615	0.858 0.800
	CAMeLBERT	0.801 0.714	0.715 0.600	0.845 0.556	0.762 1.000
PTSD	AraBERT	0.677 0.588	0.675 0.650	0.639 0.714	0.719 0.500
	CAMeLBERT	0.701 0.640	0.649 0.550	0.588 0.533	0.867 0.800
Suicidal	AraBERT	0.866 0.778	0.788 0.750	0.816 0.700	0.922 0.875
	CAMeLBERT	0.837 0.750	0.754 0.750	0.820 0.750	0.856 0.750
Bipolar	AraBERT	0.841 0.714	0.734 0.750	0.801 0.833	0.884 0.625
	CAMeLBERT	0.858 0.571	0.754 0.438	0.789 0.462	0.940 0.750
Schizophrenia	AraBERT	0.875 0.800	0.781 0.800	0.917 0.800	0.837 0.800
	CAMeLBERT	0.869 0.800	0.770 0.800	0.908 0.800	0.833 0.800
Eating dis.	AraBERT	0.748 0.727	0.614 0.700	0.776 0.667	0.722 0.800
	CAMeLBERT	0.720 0.667	0.585 0.500	0.775 0.500	0.672 1.000

Table 8: Condition keyword inventory, 341 keyword forms including English and Arabic variants, across the 15 conditions the extraction pipeline was run on. Of these, 12 are retained in the released corpus (Table 2); DID, Trichotillomania, and Paranoid PD yielded too few confirmed users and are excluded from the final corpus and downstream experiments, but are included here for completeness of the pattern description. † = restricted to suffering phrases only.

Condition	Arabic Label	Representative Keywords (Arabic / English)
Schizophrenia	الفصام	فصام، الفصام، شيزوفرينيا، اضطراب الفصام، انفصام الشخصية، الاضطراب الفصامي الوجداني، <i>schizophrenia, schizoaffective disorder</i> / سكيذوأفكتيف
Bipolar Disorder	ثنائي القطب	<i>bipolar disorder, bipolar I/II, hypomania, mania, cyclothymia</i> / ثنائي القطب، اضطراب ثنائي القطب، بايولار، هوس خفيف، مانيا، سيكلوثيميا
Depression	الاكتئاب	اكتئاب، الاكتئاب، اكتئاب حاد / مزمن، اكتئاب ما بعد الولادة، الاكتئاب الموسمي، دبريشن، <i>depression, MDD, dysthymia, postpartum depression, seasonal affective disorder</i> / سوء المزاج
Anxiety	اضطراب القلق	اضطراب القلق، قلق عام / مرضي / مزمن، رهاب اجتماعي، قلق اجتماعي، رهاب الخلاء، قلق <i>generalized anxiety, social anxiety, GAD, agoraphobia, specific phobia</i> / الانفصال
Panic Disorder	اضطراب الهلع	<i>panic attack, panic disorder</i> / نوبة هلع، نوبات هلع، الهلع، اضطراب الهلع، نوبة زعر
OCD	الوسواس القهري	وسواس قهري، الوسواس القهري، الوسواس الديني، وسواس الطهارة، ديسمورفيا، اكتناز مرضي <i>OCD, body dysmorphic disorder, BDD, hoarding disorder, scrupulosity</i>
PTSD	اضطراب ما بعد الصدمة	<i>PTSD, C-PTSD, complex PTSD, acute stress disorder</i> / اضطراب ما بعد الصدمة، صدمة نفسية، صدمة معقدة، كرب ما بعد الصدمة
Eating Disorder	اضطراب الأكل	<i>anorexia, bulimia, binge eating disorder, ARFID, orthorexia</i> / فقدان الشهية العصبي، أنوركسيا، الشره المرضي، بوليميا، نهم الطعام، <i>ARFID</i>
ADHD	فرط الحركة وتشتت الانتباه	<i>ADHD / ADHD, attention deficit hyperactivity disorder; ADD†</i> / فرط حركة، تشتت انتباه، نقص الانتباه، اضطراب الانتباه
DID	اضطراب الهوية التفارقي	<i>dissociative identity disorder, DID, multiple personality disorder</i> / اضطراب الهوية التفارقي، تعدد الشخصيات
Sleep Disorder	اضطراب النوم	<i>insomnia, sleep apnea, hypersomnia, narcolepsy</i> / الأرق، أرق، انقطاع النفس النومي، فرط النوم، الخدار
Trichotillomania	اضطراب شد الشعر	<i>trichotillomania, excoriation disorder, onychophagia</i> / شد الشعر، نتف الشعر، نتف الجلد، قضم الأظافر
BPD	اضطراب الشخصية الحدية	<i>borderline personality disorder, BPD</i> / اضطراب الشخصية الحدية، الحدية، بوردرلاين
Paranoid PD	اضطراب الشخصية الزوراني	<i>paranoid personality disorder</i> / اضطراب الشخصية الزوراني، الشخصية البارانونية، البارانونيا
Suicidal Ideation	الأفكار الانتحارية	† محاولة انتحار † انتحار / أفكار انتحارية، الفكر الانتحاري <i>suicidal thoughts, suicidal ideation; suicide attempt†, self-harm†</i> / إيذاء النفس

Table 9: Universal self-disclosure phrases by type and connector class. **B** = ب connector; **M** = من connector; **D** = direct (no connector).

Type	Conn.	Phrases
Affliction	B	أنا مصاب / أنا مصابة، أنا مريض، أنا مريضة، كشخص مصاب
Suffering	M	أعاني من / أعاني من، أنا أعاني من، أنا أعالج من، أنا أعيش مع، أصبت
Clinical Diagnosis	B	تم تشخيصي، تشفوني، تشفني، تشفني، تشفني، تشفني الطبيب، تشفني الدكتور، تشفني الطيبة، تشفني الدكتورة، الأخصائي تشفني، الأخصائية تشفني
Possessive	D	عندي، أنا عندي
Discovery /	B	طلعت مصاب، لقيت نفسي مصاب، اكتشفت إني مصاب، عرفت إن أنا مصاب، أتأكدت إني مصاب، كنت مصاب
Confirmation	D	تشخيصي هو، تشخيصي كان، طلع تشخيصي، طلع عندي، لقيت نفسي عندي، اكتشفت إن عندي، عرفت إني عندي، أتأكدت إن عندي، بعد التشخيص طلع عندي، التحليل أثبت إن عندي
Doctor Result	B	أكد الطبيب إصابتي، الدكتور أكد إصابتي، تم التأكيد على إصابتي، تم إثبات إصابتي، الدكتور أكد التشخيص، الطبيب أثبت التشخيص
	D	تشخيص الطبيب هو، تشخيص الدكتور هو، تشخيص حالي هو، تشخيص المرض هو، تم تشخيص حالي على أنها، التشخيص الذي حصلت عليه هو، طلع إن التشخيص هو، اتضح إن التشخيص كان

Table 10: Dialect-specific self-disclosure phrases. **B** = ب connector; **M** = من connector; **D** = direct.

Dialect	Conn.	Phrases
Gulf	B	أنا مصاب، مبتلي، مبتلية، ابتليت
	D	عندي، فني، أنا متعاش، جاني، جاتي
	M	أعاني من
Levantine	B	أنا مصاب
	D	عندي، معي، طلعلي، طلعتلي، صار معي، صارت معي، اجاني، اجاتي
	M	بعاني من، عم بعاني
Egyptian	B	أنا مصاب، طلعت مصاب، لقيت نفسي مصاب، اكتشفت إني مصاب، أتأكدت إني مصاب
	D	عندي، معايا، جالي، جاتي، طلعلي، طلعت عندي، لقيت نفسي عندي، اكتشفت إن عندي، أتأكدت إن عندي، عرفت إني عندي، أنا بواجه مشكلة اسمها بعاني من، اتشخصت
Maghrebi	B	أنا مصاب
	D	عندي، كاين عندي، جاني، جاتي، طلعلي، تشخصت
MSA	B	أنا مصاب، أنا مصابة، تشخصت، تشخصت
	D	لدي، تشخيصي هو، طلع تشخيصي، أتعايش مع
	M	أعاني من، تم تشخيصي

Table 11: Connector rules: phrase type, connector morpheme, keyword form generated, and example.

Phrase Type	Conn.	Keyword Form	Example
Affliction, Diagnosis, Confirmation	ب	Base kw: ب+kw ال-kw: بال+stem English: ب kw	مصاب باكتئاب مصاب بالاكئاب PTSD مصاب ب
Suffering	من	Phrase ends in من: bare kw Phrase lacks من: من+kw	أعاني من الاكتئاب عم بعاني من اكتئاب
Possessive, Discovery	Direct	Bare or definite kw	عندي اكتئاب تشخيصي هو الفصام

Table 12: Orthographic variant pairs generated automatically.

Feature	Variant A	Variant B
Hamza on alef	أنا	انا
Hamza in conjunction	إن	ان
Taa marbuta	مصابة	مصابه
Taa marbuta	مريضة	مريضه
Hamza on ا	أعاني	اعاني

	Ann-2: TP	Ann-2: FP
Ann-1: TP	172	6
Ann-1: FP	18	39

Table 13: Confusion matrix between the two human annotators (Ann-1, Ann-2) on the 235 posts both annotators labeled.

Pair	n	% agree	κ
Ann-1 vs. Ann-2	235	89.8	0.701
Ann-1 vs. Qwen	229	76.4	0.167
Ann-1 vs. Jais	227	74.9	0.171
Ann-2 vs. Qwen	224	79.0	0.138
Ann-2 vs. Jais	224	77.7	0.176
Qwen vs. Jais	218	83.0	0.231

Table 14: Pairwise agreement among all four raters (two human annotators, two LLM judges) on the 240-post sample. Per Landis and Koch's (1977) scale, human-human agreement is *substantial* ($\kappa = 0.701$) while every pair involving an LLM judge is only *slight-to-fair* ($\kappa = 0.138$ – 0.231).

Paraphrased Disagreement Examples

“Depression is my old friend, we’ve known each other for ages haha”

Ann-1: FP **Ann-2:** FP **Qwen:** TP **Jais:** TP **Likely source:** sarcasm read literally by LLMs

“I slept early last night, thank God insomnia didn’t come”

Ann-1: FP **Ann-2:** TP **Qwen:** TP **Jais:** FP **Likely source:** negation missed by Ann-2/Qwen

“My sister has been suffering from anxiety for a while”

Ann-1: FP **Ann-2:** FP **Qwen:** TP **Jais:** TP **Likely source:** third-person subject conflated with poster

“I have some obsessiveness but I’m not sure if it’s really OCD”

Ann-1: FP **Ann-2:** TP **Qwen:** TP **Jais:** TP **Likely source:** hedged/weak self-diagnosis judged differently

Table 15: Paraphrased examples of the four most frequent multi-rater disagreement patterns (analogous to the disagreement-pattern breakdown produced by our agreement-analysis script). Each example is a reconstructed illustration of the underlying phenomenon, not a verbatim CARMA post.